DualFaith at NLPCC-2026 Task 10: Faithfulness Verification for Scientific Reports

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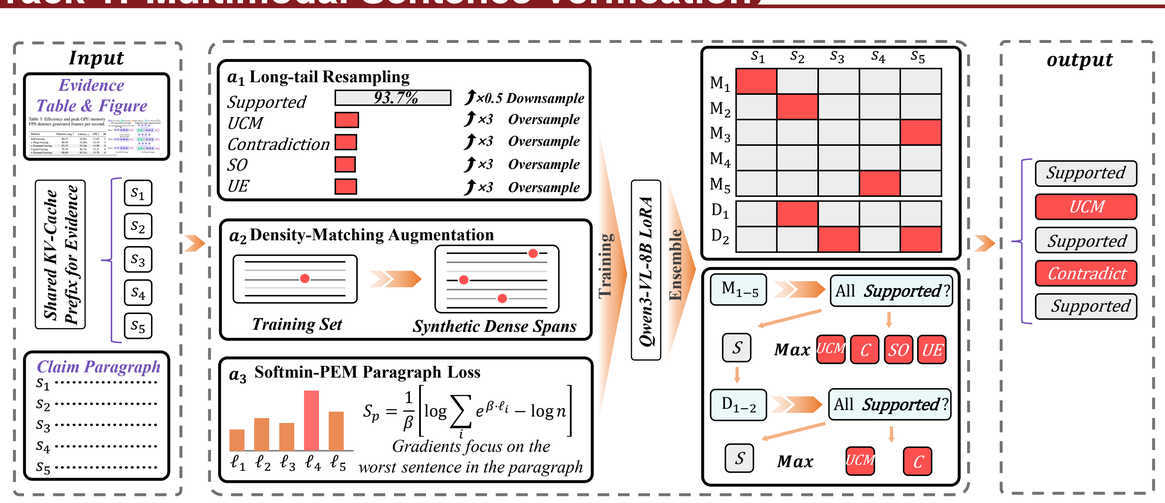
Why Scientific Faithfulness Needs Two Kinds of Evidence

AI-assisted scientific writing must be fluent *and* grounded. A results sentence may misread a figure, invent an entity, assert an unsupported mechanism, or overgeneralize a local result. A related-work claim may cite a topically relevant paper that does not actually support it. **DualFaith** addresses both failure modes with task-specific evidence modeling.

Track 1: figures & tables

Track 2: cited full text

Track 1: Multimodal Sentence Verification



Each sentence is assigned one of five labels using its claim paragraph, caption, and rendered figure/table. Qwen3-VL-8B-Instruct is adapted with LoRA; the vision tower remains frozen.

Recall-additive ensemble. Five heterogeneous members form a minority-label union. Two density-augmented members can add only causal/mechanistic or contradiction labels to positions still predicted as Supported. This directly targets the paragraph exact-match bottleneck.

LoRA $r = 16,\ \alpha = 32$ · softmin-PEM · long-tail resampling

Track 1: Data and Metric Insight

Severe imbalance: 16,438 of 17,547 train–development sentences (93.7%) are Supported; all four minority classes total only 6.3%. Only 0.46% of training paragraphs with at least six sentences contain two or more minority sentences, while approximately 84% of Phase-1 test paragraphs contain at least one. The all-Supported baseline therefore falls to 18.85.

The official score averages sentence-level Macro-F1 and paragraph exact match (PEM). Missing one minority sentence invalidates the whole paragraph, so **minority recall—not suppression—is the central Track-1 bottleneck**.

Track 1: Official Result and Sensitivity

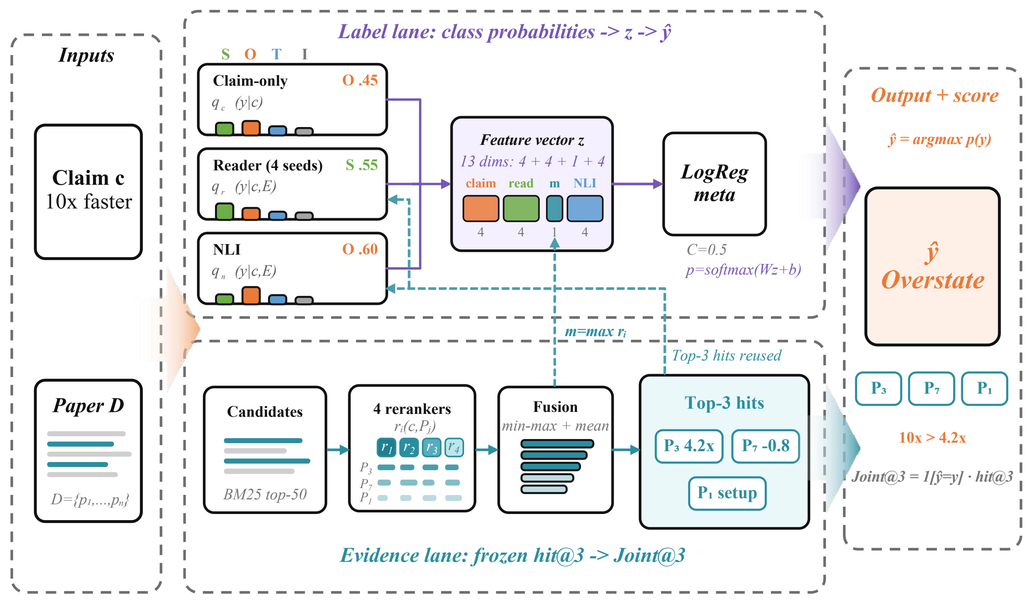
System	Final	Phase 1	Phase 2
SUAT-NLP	**68.90**	**51.85**	85.95
DualFaith	68.79	50.61	**86.97**
BIT-TrustAI	57.99	44.09	71.88
ZZUNLP_Han	55.47	42.44	68.50
YNU-HPCC	46.16	31.93	60.39

Phase-1 setting	Score	Minority predictions
Near-all Supported	18.85	46
Best single member	47.80	253
Union of 3 members	48.22	339
Union of 5 members	50.26	382
Final density-augmented union	**50.61**	**399**

Only 0.11 points behind first overall; best Phase-2 score among the top five and highest Phase-1 Macro-F1 (57.54).

Second of 11 teams

Track 2: Citation Support with Frozen Evidence



BM25 retrieves candidates; two DeBERTa cross-encoders, MonoT5-large, and frozen Qwen3-Reranker-4B are normalized per record and equally fused into a **fixed top-three evidence list**.

Domain-robust label stacking. A four-seed DeBERTa evidence reader, a claim-only classifier, a frozen multi-task NLI posterior, and the maximum evidence score feed a logistic-regression meta-classifier. Freezing retrieval isolates label calibration from evidence uncertainty.

4-way label · top-3 evidence · stacked posteriors

Track 2: Domain Shift and Metric Coupling

Phase 1 is dominated by physics/materials (64.8%), while train–development (49.3%) and Phase 2 (52.8%) are CS/AI/ML-heavy. A topic-holdout probe loses about 7 Macro-F1 points for the evidence reader but only 2 for the claim-only model.

Joint@3 succeeds only when the label is correct *and* a gold paragraph appears in the retrieved top three. Strong classification without matching evidence is insufficient; stable evidence plus robust label correction wins under shift.

Track 2: Official Result and Ablation

System	Final	Macro-F1	Joint@3	Phase 2
DualFaith	**86.27**	91.18	**81.37**	**86.33**
Lenormand	82.99	89.55	76.43	84.70
HER-DAC	81.61	93.05	70.16	81.52
ZZUNLP_Han	81.38	**94.25**	68.51	81.88
SUAT-NLP	79.01	91.02	67.00	77.13

Phase-1 label system	Score	Macro-F1	Joint@3
Four-seed reader	83.10	87.02	79.19
+ claim-only mixture	85.10	89.41	80.79
+ NLI + meta-classifier	**86.21**	**90.46**	**81.96**

A 3.28-point lead over second place. Phase 1 and blind Phase 2 differ by only 0.12, demonstrating stability across the domain shift.

First of 15 teams

Transferable Lessons

Track 1: align with error density. Higher image resolution, extra epochs, supervised chain-of-thought, OCR, and suppression rules did not yield consistent gains. Diverse union members recover complementary minority cases.

Track 2: decouple retrieval and labeling. Domain-adaptive/adversarial training, label-shift correction, extra rerankers, and eight reader seeds failed selection. Fixing evidence lets label-side signals correct domain-sensitive errors safely.

ABSTRACT IN BRIEF
DualFaith verifies whether scientific statements are supported by experimental figures and tables or by cited papers. Track 1 labels every sentence of a results paragraph against its figure; Track 2 labels an atomic claim against the cited paper and returns the supporting paragraphs. Both pipelines pair minority-recall ensembling with frozen evidence retrieval and domain-robust label stacking.

LIMITATIONS
Track 1 includes partial recipe reconstruction and PEM nondeterminism. Track 2 may inherit stylistic cues from claim-only modeling, and its development set contains only 433 records. Broader evaluation on human-written claims remains future work.

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TAKEAWAY
Faithfulness verification improves when the system is designed around the metric and the shift: add minority recall for multimodal claims, and freeze evidence before robust citation-label stacking.

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